MA4N1: Proof Structure

Universal Approximation Theorem via Stone-Weierstrass

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This project formalises the proof of the shallow-network Universal Approximation Theorem. This is proven via the Stone–Weierstrass Theorem, alongside ancillary results. This document walks through the `LEAN` files, giving a high-level overview of the proof strategies with rigour of the `LEAN` proofs.

Note: Some statements of theorems and proofs in this outline are from an AI summary of our `LEAN` code we have produced. Since this document isn't assessed, they are solely here to help readers follow the code.

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Part I

Stone-Weierstrass Theorem

To prove our primary goal (the Universal Approximation theorem), we will first re-prove the Stone-Weierstrass Theorem, as follows:

Theorem (Stone-Weierstrass)**:**
Let Ω be a compact topological space and let $\mathcal{A}$ be a subalgebra of $C(\Omega)$ that contains a non-zero constant function. Then $\mathcal{A}$ is dense in $C(\Omega)$ with the uniform norm if and only if $\mathcal{A}$ separates points in Ω.

1 Real Number Identities

Let Ω be a topological space. Let $\mathcal{A}$ be a unital subalgebra of $C(\Omega, \mathbb{R})$.

The proof begins by establishing arithmetic identities for maximum and minimum operations on real numbers using absolute values.

Lemma 1.1 (`max_id_abs`)**:**
For any $x, y \in \mathbb{R}$,

$$\max(x, y) = \frac{1}{2}(x + y + |x - y|).$$

Proof. We proceed by cases based on the ordering of real numbers:

1. **Case 1:** $x \leq y$. In this case, $\max(x, y) = y$. Also, $x - y \leq 0$, so $|x - y| = -(x - y) = y - x$. Substituting into the RHS:

   $$\frac{1}{2}(x + y + (y - x)) = \frac{1}{2}(2y) = y.$$

   The identity holds.

2. **Case 2:** $y \leq x$. In this case, $\max(x, y) = x$. Also, $x - y \geq 0$, so $|x - y| = x - y$. Substituting into the RHS:

   $$\frac{1}{2}(x + y + (x - y)) = \frac{1}{2}(2x) = x.$$

   The identity holds.

□

Lemma 1.2 (`min_id_abs`)**:**
For any $x, y \in \mathbb{R}$,

$$\min(x, y) = \frac{1}{2}(x + y - |x - y|).$$

Proof. Similar to the previous lemma, we split into cases $x \leq y$ and $y \leq x$:

1. If $x \leq y$, $\min(x, y) = x$ and $|x-y| = y-x$. The RHS becomes $\frac{1}{2}(x+y-(y-x)) = x$.
2. If $y \leq x$, $\min(x, y) = y$ and $|x-y| = x-y$. The RHS becomes $\frac{1}{2}(x+y-(x-y)) = y$.

□

2 Function Lattice Operations

We lift the real number identities to the function space $C(\Omega, \mathbb{R})$ via pointwise operations.

Lemma 2.1 (`max_id_abs_func`):
For $f, g \in C(\Omega, \mathbb{R})$, the pointwise maximum satisfies:

$$f \sqcup g = \frac{1}{2} \cdot (f + g + |f - g|).$$

Proof. By extensionality two functions are equal if they agree at all points $x \in \Omega$. At any point x, $(f \sqcup g)(x) = \max(f(x), g(x))$. Applying `max_id_abs` to the real numbers $f(x)$ and $g(x)$, we have:

$$\max(f(x), g(x)) = \frac{1}{2}(f(x) + g(x) + |f(x) - g(x)|).$$

This matches the definition of scalar multiplication and addition in the function space. □

Lemma 2.2 (`min_id_abs_func`):
For $f, g \in C(\Omega, \mathbb{R})$, the pointwise minimum satisfies:

$$f \sqcap g = \frac{1}{2} \cdot (f + g - |f - g|).$$

Proof. Analogous to the maximum case, applying `min_id_abs` pointwise at every x. □

3 Subalgebra Properties

Let $\mathcal{A}$ be a unital subalgebra of $C(\Omega, \mathbb{R})$.

Lemma 3.1 (`polynomial_comp_mem_subalg`):
Let p be a polynomial with real coefficients, and let $f \in \mathcal{A}$. Then the composition $p \circ f \in \mathcal{A}$.

Proof. We interpret $p \circ f$ as the algebraic evaluation of the polynomial p at the element f within the algebra. We proceed by induction on the structure of polynomial p:

1. **Base Case (Constants):** $p = C(r)$. The evaluation is a constant function. Since $\mathcal{A}$ is unital, it contains all scalar multiples of 1, so $C(r) \in \mathcal{A}$.

2. **Induction Step (Monomials):** If the property holds for p, we consider the monomial term corresponding to multiplying by the indeterminate X. The evaluation corresponds to $c \cdot f^n$. Since $\mathcal{A}$ is an algebra, it is closed under multiplication ($f \in \mathcal{A} \implies f^n \in \mathcal{A}$) and scalar multiplication. Thus the term is in $\mathcal{A}$.

3. **Induction Step (Addition):** If the property holds for p and q, we consider $p+q$. The map is additive: $(p+q)(f) = p(f) + q(f)$. Since $\mathcal{A}$ is a subspace, it is closed under addition. Thus $p(f) + q(f) \in \mathcal{A}$.

□

Lemma 3.2 (`exists_mem_subalg_interp_two`):
Assume $\mathcal{A}$ separates points. For any distinct $x, y \in \Omega$ and any target function $f \in C(\Omega, \mathbb{R})$, there exists $g \in \mathcal{A}$ such that:

$$g(x) = f(x) \quad \text{and} \quad g(y) = f(y).$$

Proof. Fix $x \neq y$. Since $\mathcal{A}$ separates points, there exists $h \in \mathcal{A}$ such that $h(x) \neq h(y)$. We construct g as a linear combination of shifted versions of h:

$$g = a \cdot (h - h(y)) + b \cdot (h - h(x))$$

where constants a and b are chosen to solve the interpolation system:

$$a = \frac{f(x)}{h(x) - h(y)}, \quad b = \frac{f(y)}{h(y) - h(x)}.$$

Since $h(x) - h(y) \neq 0$ (separation property), these constants are well-defined. Because $\mathcal{A}$ is a unital subalgebra, it contains constants $(h(y), h(x))$, is closed under subtraction, and closed under scalar multiplication. Thus $g \in \mathcal{A}$. Evaluating at x: The second term vanishes, leaving $a(h(x) - h(y)) = f(x)$. Evaluating at y: The first term vanishes, leaving $b(h(y) - h(x)) = f(y)$. □

4 Absolute Value Approximation

We assume Ω is a Compact Space. Let $\mathcal{A}$ be topologically closed.

Lemma 4.1 (`exists_approx_abs`):
For any $\epsilon > 0$ and any $f \in \mathcal{A}$, there exists $a \in \mathcal{A}$ such that $\|a - |f|\| < \epsilon$.

Proof.
1. **Bounding:** Let $M = \|f\|$. The image of f is contained in the compact interval $K = [-M, M]$.

2. **Continuous Absolute Value:** Let $g(y) = |y|$ for $y \in K$. This is a continuous function on a compact interval.

3. **Weierstrass Approximation:** By the classical Weierstrass Approximation Theorem for intervals, there exists a polynomial p such that $\sup_{y \in K} |p(y) - |y|| < \epsilon$.

4. **Composition:** Define $a = p \circ f$. By `polynomial_comp_mem_subalg` $f \in \mathcal{A} \implies a \in \mathcal{A}$.

5. **Verification:** For any $x \in \Omega$, $f(x) \in K$. Therefore, $|p(f(x)) - |f(x)|| < \epsilon$. Taking the sup over x, we get $\|a - |f|\| < \epsilon$.

□

Lemma 4.2 (abs_mem_closed_subalg):
If $\mathcal{A}$ is a topologically closed subalgebra, then for any $f \in \mathcal{A}$, implies $|f| \in \mathcal{A}$.

Proof. Being topologically closed means $\mathcal{A} = \overline{\mathcal{A}}$. Equivalently, $h \in \mathcal{A}$ iff for every $\epsilon > 0$, there exists $a \in \mathcal{A}$ such that $\text{dist}(a, h) < \epsilon$. By the previous lemma, for any $\epsilon > 0$, we can find $a \in \mathcal{A}$ arbitrarily close to $|f|$. Therefore, $|f| \in \overline{\mathcal{A}} = \mathcal{A}$. □

5 Closure under Pointwise Max and Min

Lemma 5.1 (sup_mem_of_closed_subalg):
If $\mathcal{A}$ is closed, then for $f, g \in \mathcal{A}$, we have $\max(f, g) \in \mathcal{A}$.

Proof. We use the identity established in Part 2:

$$\max(f, g) = \frac{1}{2}(f + g + |f - g|).$$

Since $\mathcal{A}$ is a subalgebra, $f+g \in \mathcal{A}$ and $f-g \in \mathcal{A}$. By Part 4, since $\mathcal{A}$ is closed, $|f-g| \in \mathcal{A}$. By closure under addition and scalar multiplication, the entire RHS is in $\mathcal{A}$. □

Lemma 5.2 (inf_mem_of_closed_subalg):
If $\mathcal{A}$ is closed, then for $f, g \in \mathcal{A}$, we have $\min(f, g) \in \mathcal{A}$.

Proof. Analogous to the max case, using $\min(f, g) = \frac{1}{2}(f + g - |f - g|)$. □

Lemma 5.3 (finset_sup_mem_of_closed_subalg):
Let S be a finite, non-empty index set. Let $\{F_i\}_{i \in S}$ be a family of functions in $\mathcal{A}$. If $\mathcal{A}$ is closed, then

$$\sup_{i \in S} F_i \in \mathcal{A}$$

.

Proof. We proceed by induction on the finite set S:

- **Base case (Singleton):** If $S = \{a\}$, then $\sup S = F_a$, which is in $\mathcal{A}$ by hypothesis.
- **Inductive Step (Insert):** Consider $S \cup \{a\}$. We can write:

  $$\sup_{i \in S \cup \{a\}} F_i = \max\left(F_a, \sup_{j \in S} F_j\right).$$

  By the inductive hypothesis, $F_{\text{tail}} = \sup_{j \in S} F_j \in \mathcal{A}$. Since $F_a \in \mathcal{A}$, and $\mathcal{A}$ is closed under pairwise maximum, the result follows.

□

Lemma 5.4 (finset_inf_mem_of_closed_subalg):
Let S be a finite, non-empty index set. Let $\{F_i\}_{i \in S}$ be a family of functions in $\mathcal{A}$. If $\mathcal{A}$ is closed, then

$$\inf_{i \in S} F_i \in \mathcal{A}$$

.

Proof. The proof for the infimum is identical as the above, utilizing closure under pairwise minimum. □

6 Separation of Points under Closure

Lemma 6.1 (`separatesPoints_subalg_closure`):
If $\mathcal{A}$ separates points, then its topological closure $\overline{\mathcal{A}}$ also separates points.

Proof. Let $x \neq y$. Since $\mathcal{A}$ separates points, there exists $f \in \mathcal{A}$ such that $f(x) \neq f(y)$. Since $\mathcal{A} \subseteq \overline{\mathcal{A}}$, $f \in \overline{\mathcal{A}}$. Thus, $\overline{\mathcal{A}}$ separates points. □

7 Local Upper Bound Construction

We now begin the specific construction for the Stone-Weierstrass theorem.

Lemma 7.1 (`exists_mem_local_upper_bound`):
Let $f \in C(\Omega, \mathbb{R})$, $x \in \Omega$, and $\epsilon > 0$. Assume $\mathcal{A}$ separates points and is closed. There exists $g \in \mathcal{A}$ such that:

$$g(x) = f(x) \quad \text{and} \quad \forall z \in \Omega, g(z) > f(z) - \epsilon.$$

Proof.

1. **Pointwise Interpolation:** For every $y \in \Omega$, if $y = x$, we choose the constant function $f(x)$. If $y \neq x$, we use Part 3 to find a function matching f at x and y. Using the Axiom of Choice, we obtain a family of functions $\{G_y\}_{y \in \Omega} \subset \mathcal{A}$ such that for all y:
$$G_y(x) = f(x) \quad \text{and} \quad G_y(y) = f(y).$$

2. **Open Cover:** For each y, define the set:
$$U_y = \{z \in \Omega \mid G_y(z) > f(z) - \epsilon\}.$$
Since $G_y(y) = f(y) > f(y) - \epsilon$, we have $y \in U_y$. Thus, $\{U_y\}$ covers Ω. Since G_y and f are continuous, each U_y is open.

3. **Compactness:** Since Ω is compact, extract a finite subcover corresponding to indices $S \subset \Omega$. Note that S is non-empty (assuming Ω is non-empty).

4. **Max Construction:** Define $g_x = \sup_{y \in S} G_y$. By Part 5, since each $G_y \in \mathcal{A}$ and $\mathcal{A}$ is closed, $g_x \in \mathcal{A}$.

5. **Verification:**
   - At x: For all $y \in S$, $G_y(x) = f(x)$. Thus $g_x(x) = \max(f(x), \dots) = f(x)$.
   - At arbitrary z: Since S covers Ω, $z \in U_y$ for some $y \in S$. For this specific y, $G_y(z) > f(z) - \epsilon$. Since $g_x(z)$ is the maximum over the set, $g_x(z) \geq G_y(z) > f(z) - \epsilon$.

□

8 Global Approximation (The SWT)

Lemma 8.1 (`exists_mem_global_approx`):
For any $f \in C(\Omega, \mathbb{R})$ and $\epsilon > 0$, if $\mathcal{A}$ separates points and is closed, there exists $g \in \mathcal{A}$ such that $\|g - f\| < \epsilon$.

Proof. If Ω is empty, the zero function suffices. Assume Ω is non-empty.

1. **Local Upper Bounds:** By the previous lemma, for every $x \in \Omega$, there exists $G_x \in \mathcal{A}$ such that $G_x(x) = f(x)$ and $G_x > f - \epsilon$ everywhere. Using choice, we select such a family $\{G_x\}_{x\in\Omega}$.

2. **Open Cover (Upper Condition):** We now constrain the functions from above. Define open sets:
$$V_x = \{z \in \Omega \mid G_x(z) < f(z) + \epsilon\}.$$
Since $G_x(x) = f(x) < f(x) + \epsilon$, $x \in V_x$. Thus $\{V_x\}$ covers Ω.

3. **Compactness:** Extract a finite subcover corresponding to indices S.

4. **Min Construction:** Define $g = \inf_{x\in S} G_x$. By Part 5 (finite minimum closure), $g \in \mathcal{A}$.

5. **Verification of Norm:** We must show $|g(z) - f(z)| < \epsilon$ for all z.

   - **Lower Bound:** We know $G_x(z) > f(z) - \epsilon$ for all x. Thus, the minimum $g(z) > f(z) - \epsilon$.
   - **Upper Bound:** Since S covers Ω, $z \in V_x$ for some $x \in S$. For this x, $G_x(z) < f(z) + \epsilon$. Since $g(z)$ is the minimum, $g(z) \le G_x(z) < f(z) + \epsilon$.

   Combining these, $f(z) - \epsilon < g(z) < f(z) + \epsilon$, so $\|g - f\| < \epsilon$.

□

Theorem 8.2 (Stone-Weierstrass Theorem):
Let Ω be a Compact space. Let $\mathcal{A}$ be a subalgebra of $C(\Omega, \mathbb{R})$ that separates points. Then the topological closure of $\mathcal{A}$ is the entire space $C(\Omega, \mathbb{R})$.

$$\overline{\mathcal{A}} = \top \equiv C(\Omega, \mathbb{R})$$

Proof. Let $B = \overline{\mathcal{A}}$. B is closed (by definition) and separates points (by Part 6). We wish to show $C(\Omega, \mathbb{R}) \subseteq B$. Let $f \in C(\Omega, \mathbb{R})$. By properties of the closure, $f \in B$ if and only if for every $\epsilon > 0$, there exists $g \in B$ such that $\mathrm{dist}(f, g) < \epsilon$. Applying the `exists_mem_global_approx` lemma to the algebra B, we find such a $g \in B$ satisfying $\|g - f\| < \epsilon$. Therefore, $f \in B$. □

Part II

Universal Approximation Theorem

The general form of the Universal Approximation Theorem in $\mathbb{R}$ is as follows:

Theorem (Universal Approximation in $\mathbb{R}$):
Let $\Omega \subset \mathbb{R}^d$ be a compact set. Let $\sigma : \mathbb{R} \to \mathbb{R}$ be a continuous activation function. Let $\mathcal{F}_\sigma$ be the space spanned by all functions $\varphi : \Omega \to \mathbb{R}$ of the form:

$$\varphi(x) := \sum_{i=1}^{n} a_i \sigma(w_i^T x + b_i) = a^T \sigma(Wx + b)$$

Where n is the number of neurons, $a, b \in \mathbb{R}^n$, and $W \in \mathbb{R}^{n \times d}$ with rows $w_i^T \in \mathbb{R}^{1 \times d}$.

Then $\mathcal{F}_\sigma$ is dense in $C(\Omega)$ with uniform convergence if (and only if) σ is non-polynomial.

We instead prove a lesser result for specific activation functions. We will assume that $\sigma : \mathbb{R} \to \mathbb{R}$ is an increasing (monotone) continuous function such that

$$\lim_{r\to\infty} \sigma(r) = 1 \quad \text{and} \quad \lim_{r\to-\infty} \sigma(r) = 0.$$

Theorem (The Universal Approximation theorem):
With σ as above, and $\Omega \subset \mathbb{R}^d$ a compact set, define $\mathcal{F}_\sigma$ to be the space spanned by the functions

$$\mathcal{F}_\sigma := \{\varphi_{w,b}(x) := \sigma(x^\top w + b);\ w \in \mathbb{R}^d,\ b \in \mathbb{R}\}$$

Then $\mathcal{F}_\sigma$ is dense in $C(\Omega)$ with uniform convergence. This means that for any continuous function $f : \Omega \to \mathbb{R}$ and any $\epsilon > 0$, there exists $q \in \mathbb{N}$, $w_j \in \mathbb{R}^d$ and $a_j, b_j \in \mathbb{R}$ such that

$$\sup_{x\in\Omega} \left| f(x) - \sum_{k=1}^{q} a_k \varphi_{w_k,b_k}(x) \right| \leq \epsilon.$$

This proof relies on the Stone-Weierstrass Theorem (SWT) to first prove the result for exponential neurons, and then extends this to general sigmoidal activation functions via density arguments.

1 Definitions

We begin by defining the basic components of a neural network layer.

Definition 1.1 (Domain):
Let $d \in \mathbb{N}$ be the dimension of the input space. Let $\Omega \subseteq \mathbb{R}^d$ be the domain of our functions.

Definition 1.2 (Activation Function):
Let $\sigma : \mathbb{R} \to \mathbb{R}$ be a continuous activation function.

Definition 1.3 (Single Neuron)**:**
A single neuron is a function from Ω to $\mathbb{R}$ parameterized by a weight vector $w \in \mathbb{R}^d$ and a bias $b \in \mathbb{R}$, defined as:

$$x \mapsto \sigma(\langle w, x \rangle + b)$$

Definition 1.4 (Set of Neurons)**:**
The set of all possible single neurons on Ω is:

$$\mathcal{N} = \{f \in C(\Omega, \mathbb{R}) \mid \exists w \in \mathbb{R}^d, b \in \mathbb{R}, f(x) = \sigma(\langle w, x \rangle + b)\}$$

Definition 1.5 (Shallow Neural Network $\mathcal{F}_\sigma$)**:**
$\mathcal{F}_\sigma$ is the submodule spanned by the set of neurons. Equivalently, it is the set of all finite linear combinations of neurons:

$$\mathcal{F}_\sigma = span_{\mathbb{R}}(\mathcal{N})$$

2 The Algebra of Exponential Neurons

We first analyse the specific case where the activation function is the exponential function, $\sigma(x) = \exp(x)$. We aim to show that the submodule of exponential neurons forms a subalgebra.

Lemma 2.1 (`neuron_mul_neuron_exp`)**:**
The product of two exponential neurons is an exponential neuron. Specifically:

$$neuron(w_1, b_1) \cdot neuron(w_2, b_2) = neuron(w_1 + w_2, b_1 + b_2).$$

Proof. By extensionality, we examine the value at any $x \in \Omega$:

$$\begin{aligned}
\exp(\langle w_1, x \rangle + b_1) \cdot \exp(\langle w_2, x \rangle + b_2) &= \exp((\langle w_1, x \rangle + b_1) + (\langle w_2, x \rangle + b_2)) \\
&= \exp((\langle w_1, x \rangle + \langle w_2, x \rangle) + (b_1 + b_2)) \\
&= \exp(\langle w_1 + w_2, x \rangle + (b_1 + b_2))
\end{aligned}$$

This is exactly the definition of the neuron with weights $w_1 + w_2$ and bias $b_1 + b_2$. $\square$

Lemma 2.2 (`one_mem_F_exp`)**:**
The constant function 1 belongs to the span of exponential neurons.

Proof. Consider the neuron with $w = 0$ and $b = 0$. For any x:

$$\exp(\langle 0, x \rangle + 0) = \exp(0) = 1.$$

Since this is a valid neuron, it is in the span $\mathcal{F}_\sigma$. $\square$

Lemma 2.3 (`mul_mem_F_exp`)**:**
The submodule of exponential neurons is closed under multiplication. That is, if $f, g \in \mathcal{F}_\sigma(\exp)$, then $f \cdot g \in \mathcal{F}_\sigma(\exp)$.

Proof. We proceed by induction on the structure of the submodule span:

1. **Base Case (Neurons):** If f and g are single neurons, their product is a single neuron (by the first lemma above), which is in the span.
2. **Zero Case:** If $g = 0$, $f \cdot 0 = 0$, which is in the submodule.
3. **Additivity:** If $g = g_1 + g_2$, then $f(g_1 + g_2) = fg_1 + fg_2$. By the inductive hypothesis, fg_1 and fg_2 are in the span, so their sum is too.
4. **Homogeneity:** If $g = r \cdot g'$, then $f(rg') = r(fg')$. By hypothesis fg' is in the span, so its scalar multiple is too.

Repeating this induction for f completes the proof. □

Definition 2.4 (Subalgebra $\mathcal{F}_{\text{exp}}$)**:**
We define $\mathcal{F}_{\text{exp}}$ as the subalgebra of $C(\Omega, \mathbb{R})$ generated by exponential neurons. Its carrier set is the submodule $\mathcal{F}_\sigma(\text{exp})$, and the algebraic operations are justified by the lemmas above.

3 Separation of Points

To apply the Stone-Weierstrass Theorem, we must show that $\mathcal{F}_{\text{exp}}$ separates points.

Lemma 3.1 (`F_exp_separates_points`)**:**
For any distinct $x, y \in \Omega$, there exists $f \in \mathcal{F}_{\text{exp}}$ such that $f(x) \neq f(y)$.

Proof. Let $x \neq y$. Define vector $v = x - y$. Since $x \neq y$, $v \neq 0$ and $\|v\|^2 > 0$. We construct a specific neuron with:

$$w = \frac{x - y}{\|x - y\|^2}, \quad b = -\langle w, y \rangle.$$

We check the values at y and x: 1. At y: $\langle w, y \rangle + b = \langle w, y \rangle - \langle w, y \rangle = 0$. Thus $f(y) = \exp(0) = 1$. 2. At x:

$$\begin{aligned} \langle w, x \rangle + b &= \langle w, x \rangle - \langle w, y \rangle = \langle w, x - y \rangle \\ &= \langle \frac{v}{\|v\|^2}, v \rangle = \frac{\langle v, v \rangle}{\|v\|^2} = \frac{\|v\|^2}{\|v\|^2} = 1. \end{aligned}$$

Thus $f(x) = \exp(1) = e$. Since $e \neq 1$, $f(x) \neq f(y)$. Since f is a neuron, $f \in \mathcal{F}_{\text{exp}}$. □

4 UAT for Exponential Neurons

We now assume Ω is a **Compact Space**.

Theorem 4.1 (`F_exp_UAT_top`)**:**
The subalgebra $\mathcal{F}_{\text{exp}}$ is dense in $C(\Omega, \mathbb{R})$.

$$\overline{\mathcal{F}_{\text{exp}}} = \top$$

Proof. We apply the **Stone-Weierstrass Theorem** (formalized in `SWT.lean`). The subalgebra $\mathcal{F}_{\text{exp}}$ is unital and separates points (Part 3). Since Ω is compact Hausdorff (as a subspace of Euclidean space), SWT applies. Therefore, the topological closure of $\mathcal{F}_{\text{exp}}$ is the entire space of continuous functions. □

Corollary 4.2 (F_exp_UAT)**:**
For any $f \in C(\Omega, \mathbb{R})$ and $\epsilon > 0$, there exists $g \in \mathcal{F}_{\text{exp}}$ such that $\|g - f\| < \epsilon$.

5 Approximation by Sigmoidal Functions

We now turn to the general case where σ is not exp, but a sigmoidal function.

Definition 5.1 (Sigmoidal)**:**
A function $\sigma : \mathbb{R} \to \mathbb{R}$ is Sigmoidal if: 1. σ is Monotone. 2. $\lim_{t\to-\infty} \sigma(t) = 0$. 3. $\lim_{t\to+\infty} \sigma(t) = 1$.

Assume σ is Sigmoidal and continuous. Let $T \subset \mathbb{R}$ be a compact subset.

Definition 5.2 (Scalar Neurons & $\mathcal{H}_\sigma$)**:**
Let T be a compact subset of $\mathbb{R}$. The set of scalar neurons on T is $\{f(t) = \sigma(ut + v) \mid u, v \in \mathbb{R}\}$. $\mathcal{H}_\sigma(T)$ is the submodule spanned by these scalar neurons.

Lemma 5.3 (`sigma_approx_step`)**:**
The sigmoidal function can approximate a step function. For any $z \in \mathbb{R}, \epsilon > 0, \delta > 0$, there exist $w, b \in \mathbb{R}$ such that $\sigma(wt + b)$ is within ϵ of 0 for $t < z - \delta$ and within ϵ of 1 for $t > z + \delta$.

Proof. Since $\sigma \to 0$ at $-\infty$ and $\sigma \to 1$ at $+\infty$, there exist bounds M_1, M_2 where σ is ϵ-close to its limits. We choose a large scaling factor w and bias b to compress the transition of σ into the interval $(z - \delta, z + \delta)$ and shift it to z.

$$w = \frac{|M_1| + |M_2| + 1}{\delta}, \qquad b = -wz$$

The proof checks the inequalities for $t < z - \delta$ and $t > z + \delta$ using the monotonicity of the affine transformation. □

Lemma 5.4 (`exp_in_H_sigma`)**:**
The exponential function on T, denoted $ext_T(t) = \exp(t)$, can be approximated by $\mathcal{H}_\sigma$. For any $\epsilon > 0$, there exists $h \in \mathcal{H}_\sigma(T)$ such that $\|h - ext_T\| < \epsilon$.

Remark 5.5:
The formal proof for this lemma is omitted in the code (`sorry`). This is because the proof of this approximation requires quite fiddely constructions of step functions for the exponential. It is a lot easier to see this mathematically than it is in `LEAN`, and the full proof is outlined in the notes from MA3K1 - Maths of Machine Learning (Part 3).

Proof. 1. Since T is compact, the function $f(t) = \exp(t)$ is uniformly continuous on T. There exists a $\gamma > 0$ such that for any $|x - y| < \gamma$, $|f(x) - f(y)| < \epsilon/2$.

2. Partition the interval T into N points $x_0, x_1, \ldots, x_N$ such that the distance between points is less than γ. We can approximate $f(x)$ with a step function $S_N(x)$:

$$S_N(x) = f(x_0) + \sum_{j=1}^{N} (f(x_j) - f(x_{j-1})) \cdot \mathbb{1}_{[x_j,\infty)}(t)$$

. Because f is increasing, the coefficients $(f(x_j) - f(x_{j-1}))$ are always positive.

3. Use `sigma_approx_step` to replace each indicator function $\mathbb{1}_{[x_j,\infty)}$ with a sigmoid function $\sigma(w_j x + b_j)$. Choose w_j, b_j large enough so that $\sigma(w_j x + b_j)$ is within a tiny error η of the step function outside a small transition window $(x_j - \delta, x_j + \delta)$

4. Define the neural network $h(t) \in H_\sigma$ as:

$$h(t) = f(x_0) + \sum_{j=1}^{N} (f(x_j) - f(x_{j-1})) \cdot \sigma(w_j x + b_j)$$

5. Outside the transition windows, the sigmoids are essentially 0 or 1, so $h(x) \approx S_N(x)$. Since $|f(x) - S_N(x)| < \epsilon/2$, by construction.

6. Inside the transition windows, both $f(x)$ and $h(x)$ are monotone increasing, so their values inside the small window $(x_j - \delta, x_j + \delta)$ are "trapped" between their values at the endpoints. Since the window width $< 2\delta$, the value of f doesn't change by more than ϵ. Therefore, the error cannot "spike" inside the transition region.

□

6 Lifting to $\mathcal{F}_\sigma$

We link the 1D approximation $\mathcal{H}_\sigma$ back to the high-dimensional space $\mathcal{F}_\sigma$.

Lemma 6.1 (`lift_H_sigma_to_F_sigma`):
Let $L \in C(\Omega, T)$ be an affine map $L(x) = \langle w, x\rangle + b$. If $h \in \mathcal{H}_\sigma(T)$ is an approximation function, then the composition $h \circ L$ belongs to $\mathcal{F}_\sigma$.

Proof. We invoke induction on the span of $\mathcal{H}_\sigma$. The base case is a scalar neuron $h(t) = \sigma(ut + v)$. The composition is:

$$(h \circ L)(x) = \sigma(u(\langle w, x\rangle + b) + v) = \sigma(\langle uw, x\rangle + (ub + v)).$$

This is a standard neuron in Ω with weights uw and bias $ub + v$, hence in $\mathcal{F}_\sigma$. Linearity handles the sum and scalar multiplication cases. □

Lemma 6.2 (`exp_neuron_approx_in_F_sigma`):
Any exponential neuron $f(x) = \exp(\langle w, x\rangle + b)$ can be approximated by functions in $\mathcal{F}_\sigma$.

Proof. 1. Let $L(x) = \langle w, x\rangle + b$. Since Ω is compact, the range $T = L(\Omega) \subset \mathbb{R}$ is compact.

2. By `exp_in_H_sigma`, there exists $h \in \mathcal{H}_\sigma(T)$ approximating exp on T within ϵ.

3. Define $G = h \circ L$. By the lifting lemma, $G \in \mathcal{F}_\sigma$.

4. The error is:

$$\|G - f\|_\Omega = \sup_{x \in \Omega} |h(L(x)) - \exp(L(x))| = \sup_{t \in T} |h(t) - \exp(t)| < \epsilon.$$

□

7 UAT for Continuous Sigmoidal Activation Functions

We combine the results to prove the general UAT.

Lemma 7.1 (`F_exp_subset_closure_F_sigma`)**:**
The algebra of exponential neurons is contained in the topological closure of the sigmoidal networks:

$$\mathcal{F}_{\exp} \subseteq \overline{\mathcal{F}_\sigma}.$$

Proof. Let $f \in \mathcal{F}_{\exp}$. By definition, f is a linear combination of exponential neurons. By the previous lemma, every exponential neuron can be approximated arbitrarily well by elements of $\mathcal{F}_\sigma$, meaning every exponential neuron is in $\overline{\mathcal{F}_\sigma}$. Since $\overline{\mathcal{F}_\sigma}$ is a subspace (closure of a subspace), it contains all linear combinations of its elements. Thus $f \in \overline{\mathcal{F}_\sigma}$. □

Theorem 7.2 (`monotone_UAT_top`)**:**
Let σ be a continuous, sigmoidal activation function. Then $\mathcal{F}_\sigma$ is dense in $C(\Omega, \mathbb{R})$.

$$\overline{\mathcal{F}_\sigma} = \top = C(\Omega, \mathbb{R})$$

Proof. We have the chain of inclusions:

$$\top = \overline{\mathcal{F}_{\exp}} \subseteq \overline{\overline{\mathcal{F}_\sigma}} = \overline{\mathcal{F}_\sigma} \subseteq \top$$

1. $\overline{\mathcal{F}_{\exp}} = \top$: Established in Part 4 (UAT for Exp).

2. $\mathcal{F}_{\exp} \subseteq \overline{\mathcal{F}_\sigma}$: Established in the lemma above.

3. Closure is monotonic: $A \subseteq B \implies \overline{A} \subseteq \overline{B}$.

Thus, $\overline{\mathcal{F}_\sigma} = \top = C(\Omega, \mathbb{R})$. □

Theorem 7.3 (`monotone_UAT`)**:**
For any continuous function $f \in C(\Omega, \mathbb{R})$ and $\epsilon > 0$, there exists a shallow neural network $g \in \mathcal{F}_\sigma$ such that $\|g - f\| < \epsilon$.

Proof. By the density theorem above, f is in the closure of $\mathcal{F}_\sigma$. The definition of closure in a metric space implies the existence of such a g. □

Part III

Source Material (MA3K1)

The inspiration for our approach to proving this form of the Universal Approximation Theorem comes from the MA3K1 - Maths of Machine Learning Module. Below is the section that we referred to:

3.1.1 Universal approximation

We will assume that $\sigma : \mathbb{R} \to \mathbb{R}$ is a non-decreasing continuous function such that

$$\lim_{r\to\infty} \sigma(r) = 1 \quad \text{and} \quad \lim_{r\to-\infty} \sigma(r) = 0. \tag{1}$$

Theorem 3.4 (The Universal Approximation theorem)**:**
Suppose that σ satisfies (1). For any compact set $\Omega \subset \mathbb{R}^p$, the space spanned by the functions

$$\mathcal{F}_\sigma := \{\varphi_{w,b}(x) := \sigma(x^\top w + b);\ w \in \mathbb{R}^p,\ b \in \mathbb{R}\}$$

is dense in $\mathcal{C}(\Omega)$ with uniform convergence. This means that for any continuous function $f : \Omega \to \mathbb{R}$ and any $\epsilon > 0$, there exists $q \in \mathbb{N}$, $w_j \in \mathbb{R}^p$ and $a_j, b_j \in \mathbb{R}$ such that

$$\sup_{x\in\Omega} \left| f(x) - \sum_{k=1}^{q} a_k \varphi_{w_k,b_k}(x) \right| \leq \epsilon.$$

To prove this result, we will make use of a fundamental result in approximation theory:

Theorem 3.5 (The Stone-Weierstrass Theorem)**:**
Let $\Omega \subset \mathbb{R}^p$ be compact and let $\mathcal{F}$ be a subalgebra of $\mathcal{C}(\Omega)$: that is,

- $1 \in \mathcal{F}$
- $f \cdot g \in \mathcal{F}$
- $\alpha f + \beta g \in \mathcal{F}$ *for all* $\alpha, \beta \in \mathbb{R}$.

Assume also that $\mathcal{F}$ separates points in Ω: for all $x \neq y$ there exists $f \in \mathcal{F}$ such that $f(x) \neq f(y)$. Then, $\mathcal{F}$ is dense in $\mathcal{C}(\Omega)$ with uniform norm: for all continuous $g : \Omega \to \mathbb{R}$ and $\epsilon > 0$, there exists $f \in \mathcal{F}$ such that $\|f - g\|_\infty \leq \epsilon$.

Remark 3.6:
The separation of points is necessary: if x, x' cannot be separated, then we can define a continuous function such that $g(x) \neq g(x')$ but there will be no function $f \in \mathcal{F}$ that can approximate g. E.g., $g(z) = \langle z - x, x' - x\rangle$. Clearly $g(x) = 0$ and $g(x') = \|x - x'\|^2 \neq 0$.

Lemma 3.7:
For all $g \in \mathcal{C}(\Omega)$ and all $\epsilon > 0$, there exists $f \in \operatorname{Span} \mathcal{F}_{\exp}$ such that $\|f - g\|_\infty \leq \epsilon$.

Proof. We simply need to verify that the conditions of Theorem 3.5 hold for $\operatorname{Span}\mathcal{F}_{\exp}$.

- Clearly, all elements of $\mathcal{F}_{\exp}$ are continuous functions.
- Observe that $x \mapsto \exp(\langle 0, x\rangle) = 1$ is in $\mathcal{F}_{\exp}$.
- $\operatorname{Span}\mathcal{F}_{\exp}$ is a linear subspace and this is an algebra by the property that $\exp(s+t) = \exp(s)\exp(t)$.
- Separation of points: if $x \neq y$, let $h(z) = \exp(\langle x-y, z-y\rangle/\|x-y\|^2)$. Then, $\exp(1) = h(x) \neq h(y) = 1$.

□

Proof of Theorem 3.4. We are now ready to prove Theorem 3.4. Given any $g \in \mathcal{C}(\Omega)$, by the universal approximation property of $\operatorname{Span}\mathcal{F}_{\exp}$ (Lemma 3.7), there exists $h \in \operatorname{Span}\mathcal{F}_{\exp}$ such that $\|h-g\|_\infty < \epsilon/2$. That is, we can find $h(x) = \sum_{j=1}^{q} a_j \exp(x^\top w_j + b_j)$ such that $|g(x) - h(x)| < \epsilon/2$ for all $x \in \Omega$.

It remains to show that $t \in \mathbb{R} \mapsto \exp(t)$ can be arbitrarily approximated by functions in $\operatorname{Span}\mathcal{F}_\sigma$ on a compact domain. We then simply replace every exp in h by some element $\sum_l b_l\sigma(u_l \cdot + v_l) \in \operatorname{Span}\mathcal{F}_\sigma$. Indeed, suppose that for any compact domain T, for all $\eta > 0$ there exists p such that

$$\sup_{t\in T} |\exp(t) - p(t)| \leq \eta$$

for some $p(t) = \sum_{l=1}^{L} b_l\sigma(u_l t + v_l)$. If we simply replace every exp in h by p, then

$$\tilde{h}(x) = \sum_{j=1}^{q} a_j p(x^\top w_j + b_j) = \sum_{j=1}^{q} a_j \sum_{l=1}^{L} b_l\sigma(u_l x^\top w_j + b_j + v_l) \in \operatorname{Span}\mathcal{F}_\sigma$$

and

$$|g(x) - \tilde{h}(x)| \leq |g(x) - h(x)| + |h(x) - \tilde{h}(x)| \leq \frac{\epsilon}{2} + \sum_{j=1}^{q} |a_j| \|p - \exp\|_\infty < \epsilon$$

if we choose $\eta = \frac{\epsilon}{2\|a\|_1}$.

Precisely, let f be a continuous function on $[0,1]$ and it is therefore uniformly continuous. The key observation is that any $h \in \operatorname{Span}(\mathcal{F}_\sigma)$ can be viewed as a linear combination of smoothed step functions. Observe that as $\gamma \to 0$, $h_\gamma(x) := \varphi_{w/\gamma, b/\gamma}(x) \to 1_{[-b/w,\infty)}(x)$. This implies that

$$\sum_{k=1}^{q} a_k \varphi_{w_k/\gamma, b_k/\gamma}(x) \to \sum_{k=1}^{q} a_k 1_{[-b_k/w_k,\infty)}(x).$$

The result follows because f can be arbitrarily approximated by step functions on $[0,1]$. In particular, for all $\epsilon > 0$, there exists N such that $|f(x) - f(x')| \leq \epsilon$ for all $|x - x'| \leq 1/N$. Take

$$h(x) = \sum_{j=1}^{N} f(j/N) 1_{[(j-1)/N, j/N)}(x).$$

Then $|f(x) - h(x)| \leq |f(x) - f(j/N)| < \epsilon$ for some j such that $|j/N - x| \leq 1/N$.

To make this precise, let $w > 0$. If $x < -b/w$ implies $\frac{w}{\gamma}x + \frac{b}{\gamma} < 0$, then $\lim_{\gamma\to 0}\sigma(\frac{w}{\gamma}x + \frac{b}{\gamma}) = 0$. Similarly, if $x > -b/w$, $\lim_{\gamma\to 0}\sigma(\frac{w}{\gamma}x + \frac{b}{\gamma}) = 1$. Combining this with monotonicity of σ, for all $z \in \mathbb{R}$, $\epsilon, \delta > 0$ there exists w, b such that

$$|\sigma(wx + b) - 1_{[z,\infty)}(x)| < \epsilon, \quad \forall x \notin (z - \delta, z + \delta). \tag{2}$$

Now, $f(x) = \exp(x)$ is monotone. Let N be such that $|f(x) - f(x')| < \epsilon/2$ whenever $|x - x'| < 1/N$. So,

$$S_N := \sum_{j=0}^{N-1} f(t_j)1_{[t_j,t_{j+1})} = f(t_0)1_{[t_0,\infty)} + \sum_{j=1}^{N-1}(f(t_j) - f(t_{j-1}))1_{[t_j,\infty)}$$

on $[0, 1]$ satisfies $\|f - S_N\|_\infty < \epsilon/2$. Let $t_j = j/N$, $\delta < 1/(4N)$, $\hat{w}_j, \hat{b}_j$ be such that $t_j = -\hat{b}_j/\hat{w}_j$, and $w_j = \hat{w}_j/\gamma$, $b_j = \hat{b}_j/\gamma$. From (2), $\varphi_j(x) = \sigma(w_jx + b_j)$ approximates $1_{[t_j,\infty)}$ up to $\epsilon/(2\|f\|_\infty)$ in uniform norm outside of $(t_j - \delta, t_j + \delta)$ by choosing γ sufficiently small. So,

$$h(x) = \sum_{j=1}^{N-1}(f(t_j) - f(t_{j-1}))\varphi_j + f(t_0)\varphi_0,$$

satisfies $|h(x) - S_N(x)| < \epsilon/2$ for all $x \notin \cup_j(t_j - \delta, t_j + \delta)$.

Moreover, h is monotone since $x \mapsto \sigma(wx+b)$ is monotone and $f(x) = \exp(x)$ is monotone. For $x \in (t_j - \delta, t_j + \delta)$,

$$\begin{aligned}
|f(x) - h(x)| &\leq \max(|f(x) - h(t_j - \delta)|, |f(x) - h(t_j + \delta)|) \quad \text{since } h \text{ monotone} \\
&\leq \max(|f(x) - f(t_j - \delta)|, |f(x) - f(t_j + \delta)|) + \epsilon \\
&< 2\epsilon \quad \text{as } f \text{ is unif. cont.}
\end{aligned}$$

□